IZZY GROSOF

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RESEARCH INTERESTS

Optimization and performance analysis of stochastic scheduling models for modern computing systems, especially multiserver systems, multiserver jobs, tail latency, scheduling with predictions, and reinforcement learning for queueing.

ACADEMIC EMPLOYMENT

9/2024 -: Assistant Professor (Tenure-track) at Northwestern University, Department of Industrial Engineering and Management Science.

2/2024 - 8/2024: Postdoctoral Fellow at University of Illinois at Urbana-Champaign, advised by Prof. R. Srikant & Prof. Siva Theja Maguluri.

8/2023 - 1/2024: Tennenbaum Postdoctoral Fellow at Georgia Tech, advised by Prof. Siva Theja Maguluri & Prof. R. Srikant.

EDUCATION

2017 - 2023: PhD in Computer Science. Carnegie Mellon University, Pittsburgh, PA.
Thesis title: "Optimal Scheduling in Multiserver Queues". Advisor: Prof. Mor Harchol-Balter.
Winner of ACM SIGMETRICS 2023 Doctoral Dissertation Award

2013 - 2017: M.E. and B.S. in Computer Science. Massachusetts Institute of Technology, Cambridge, MA. GPA 4.96/5
Master's Thesis in information-theoretic cryptography: "Secure communication: CDS, PIR, PSM"
Master's Advisor: Prof. Vinod Vaikunatanathan
Bachelor's Advisor: Prof. Frans Kaashoek

ADVISING

Ziyuan Wang: IEMS PhD student. Began advising Spring 2024. Proposal: Spring 2025.
Cameron Curtis: IEMS PhD student, co-advised with Prof. Seyed Irvani. Began advising Winter 2024. Proposal: Spring 2026.
Heyuan Yao: IEMS PhD student. Began advising Spring 2025.

Sianna Perez: Undergraduate student. Summer 2026-ongoing.
Daniela Chang Kim: Undergraduate student. Summer Undergraduate Research Grant (SURG) recipient. Summer 2026-ongoing.
Jooyoung Oh: Undergraduate student. SURG recipient. Summer 2026-ongoing.
Bryan Yuen: Undergraduate student. URAP recipient. Fall 2025-Spring 2026.
Yanis Zain: Undergraduate student. Summer 2025.
Willow Kowalik: Undergraduate student. Winter-Spring 2025.

HONORS

2026 INFORMS JFIG Paper Competition finalist
2025 Cole-Higgins Award for Excellence in Teaching from the McCormick School of Engineering at Northwestern
2025 INFORMS Graduate Teaching Award from the IEMS Department at Northwestern
ACM SIGMETRICS 2023 Doctoral Dissertation Award
INFORMS 2022 George Nicholson Award
2023 Siebel Scholar
ACM SIGMETRICS 2021 Best Paper Award
ACM SIGMETRICS 2019 Best Student Paper Award
IFIP Performance 2018 Best Student Paper Award
ACM SIGMETRICS 2023 Student Research Competition Winner

REFEREED PUBLICATIONS – JOURNAL PAPERS

Link to my publications
Note: The primary name I use is Izzy, and I have also published under Isaac.
Bold text indicates students I advise.

Ziyuan Wang and Izzy Grosof. "Pareto-Optimal Scheduling in the Half-batch Multiserver-job Model". Accepted to *Performance Evaluation* as of June 2026. Conference version accepted to *IFIP Performance*.

Céline Comte and Isaac Grosof. "Graph-Based Product Form". *Queueing Systems*, June 2026.
Note: Joint first authors.

Isaac Grosof and Hayriye Ayhan. "Multiserver-job Response Time under Multilevel Scaling". *Probability in the Engineering and Information Sciences*, March 2026.

Izzy Grosof and Daniela Hurtado-Lange. "Outperforming Multiserver SRPT at All Loads". Accepted to *ACM SIGMETRICS* as of December 2025. Under expedited review at a journal as of January 2026. Received minor revision, resubmitted.

Zhongrui Chen, Adityo Anggraito, Diletta Olliaro, Andrea Marin, Marco Ajmone Marsan, Benjamin Berg, Izzy Grosof. "Improving Nonpreemptive Multiserver Job Scheduling with Quickswap". *Performance Evaluation*, March 2026. Conference version at *IFIP Performance*, CWI, Amsterdam, NL, November 2025.

Isaac Grosof, Siva Theja Maguluri, R. Srikant. "Convergence for Natural Policy Gradient for a Family of Infinite-State Queueing MDPs". *Queueing Systems*, August 2025.

Zhongrui Chen, Isaac Grosof, Benjamin Berg. "Improving Multiresource Job Scheduling with Markovian Service Rate Policies." *Proceedings of the ACM Measurement and Analysis of Computer Systems – SIGMETRICS*. Conference version at *ACM SIGMETRICS*, Long Island, NY, June 2025.

Runhan Xie, Isaac Grosof, Ziv Scully. "Heavy-Traffic Optimal Size- and State-Aware Dispatching." *Proceedings of the ACM Measurement and Analysis of Computer Systems – SIGMETRICS*: Volume 8, Number 1, Article 9 (March 2024), pp. 9:1 – 9:36, June 2024, Venice, Italy. Conference version at *ACM SIGMETRICS/IFIP Performance*, June 2024.

Isaac Grosof, Yige Hong, Mor Harchol-Balter, Alan Scheller-Wolf. "The RESET and MARC Techniques, with Application to Multiserver-Job Analysis." *Performance Evaluation*, vol. 162, 2023, pp. 102378 [sic].
Conference version at *IFIP Performance 2023*, Chicago, IL, November 2023.
A short version of this paper, entitled "The RESET Technique for Multiserver-Job Analysis," was the **winner of the ACM SIGMETRICS 2023 Student Research Competition**.

Isaac Grosof, Ziv Scully, Mor Harchol-Balter, Alan Scheller-Wolf. "Optimal Scheduling in the Multiserver-job Model under Heavy Traffic." *Proceedings of the ACM Measurement and Analysis of Computer Systems – SIGMETRICS*: Volume 6, Number 3, Article 51 (Sigmetrics 2023), pp. 51:1 – 51:32, December 2022.

Isaac Grosof, Mor Harchol-Balter, Alan Scheller-Wolf. "WCFS: A new framework for analyzing multiserver systems." *Queueing Systems*, Volume 102, pp. 143-174. July 2022.

Isaac Grosof, Kunhe Yang, Ziv Scully, Mor Harchol-Balter. "Nudge: Stochastically improving upon FCFS." *Proceedings of the ACM Measurement and Analysis of Computer Systems – SIGMETRICS*: Volume 5, Number 2, Article 99 (2021), pp. 99:1 – 99:25, June 2021, Beijing, China.
Winner of ACM SIGMETRICS 2021 Best Paper Award

Ziv Scully, Isaac Grosof and Mor Harchol-Balter. "The Gittins Policy is Nearly Optimal in the M/G/k under Extremely General Conditions." *Proceedings of the ACM Measurement and Analysis of Computer Systems – SIGMETRICS*: Volume 4, Number 3, Article 43 (Dec 2020), pp. 43:1 – 43:29, June 2021, Beijing, China.
Winner of INFORMS 2022 George Nicholson Award

Ziv Scully, Isaac Grosof, Mor Harchol-Balter. "Optimal Multiserver Scheduling with Unknown Job Sizes in Heavy Traffic." *Performance Evaluation*, vol. 145, 2021, pp. 1-31.
Conference version appeared in *38th International Symposium on Computer Performance, Modeling, Measurements, and Evaluation (Performance 2020)*, Milan, Italy, November 2020.

Isaac Grosof, Ziv Scully, Mor Harchol-Balter. "Load Balancing Guardrails: Keeping Your Heavy Traffic on the Road to Low Response Times." *Proceedings of the ACM Measurement and Analysis of Computer Systems – SIGMETRICS*: Volume 3, Issue 2, Article 42 (June 2019), pp. 42:1 – 42:31, 2019.
Conference version appeared in *Proceedings of ACM Sigmetrics/Performance 2019 Conference on Measurement and Modeling of Computer Systems (SIGMETRICS 19)*, Pheonix, AZ. June 2019.
Winner of ACM SIGMETRICS 2019 Best Student Paper Award
Invited mini-plenary paper at ACM STOC 2021 TheoryFest

Isaac Grosof, Ziv Scully, Mor Harchol-Balter. "SRPT for Multiserver Systems." *Performance Evaluation*, vol. 127-128, Nov. 2018, pp. 154-175.
Conference version appeared in *36th International Symposium on Computer Performance, Modeling, Measurements, and Evaluation (Performance 2018)*, Toulouse, France, December 2018.
Winner of IFIP Performance 2018 Best Student Paper Award.

REFEREED PUBLICATIONS – FULL-LENGTH (≈ 20 PAGE) CONFERENCE PAPERS

Ziv Scully, Isaac Grosof, Michael Mitzenmacher. "Uniform Bounds for Scheduling with Job Size Estimates." *13th Innovations in Theoretical Computer Science Conference (ITCS 2022)*: Volume 215, Article 114 (2022), pp. 114:1 – 114:30, January 2022.
Invited Paper at ACM STOC 2022: Algorithms with Predictions

Ben Berg, Daniel Berger, Sara McAllister, Isaac Grosof, Sathya Gunasekar, Jimmy Lu, Michael Uhlar, Jim Carrig, Nathan Beckmann, Mor Harchol-Balter, Greg Ganger. "The CacheLib Caching Engine: Design and Experiences at Scale." 14th USENIX Symposium on Operating Systems Design and Implementation (OSDI 2020), Banff, Canada, November 2020.

Erik D. Demaine, Isaac Grosof, Jayson Lynch, and Mikhail Rudoy. "Computational Complexity of Motion Planning of a Robot through Simple Gadgets." *Ninth International Conference on Fun with Algorithms.* La Maddalena, Italy. 2018.

Erik D. Demaine, Isaac Grosof, and Jayson Lynch. "Push-Pull Block Puzzles are Hard." *International Conference on Algorithms and Complexity.* Athens, Greece. 2017.

REFEREED PUBLICATIONS – SHORT (≤ 10 PAGE) CONFERENCE PAPERS

Ziyuan Wang, Izzy Grosof. "Product Forms in Closed Order Independent Queues". Accepted to the MAMA workshop at SIGMETRICS 2026.

Zhongrui Chen, **Heyuan Yao**, Izzy Grosof, Benjamin Berg. "Non-preemptive Datacenter Scheduling via Scaling Cycles". Accepted to the MAMA workshop at SIGMETRICS 2026.

Isaac Grosof, Mor Harchol-Balter. "ServerFilling: A better approach to packing multiserver jobs." *Proceedings of the 5th workshop on Advanced tools, programming languages, and platforms for implementing and evaluating algorithms for distributed systems (ApPLIED 2023).* In conjunction with PODC 2023 Conference. Article No. 7, pp. 1–5, Orlando, CA, June 2023.

Isaac Grosof, Mor Harchol-Balter, Alan Scheller-Wolf. "New Stability Results for Multiserver-job Models via Product-form Saturated Systems." *Workshop on Mathematical Performance Modeling and Analysis (MAMA 2023).* Orlando, FL, June 2023.

Isaac Grosof, Naifeng Zhang, Marijn Heule. "Towards the shortest DRAT proof of the Pigeonhole Principle." *Pragmatics of SAT.* In conjunction with SAT 2022. August 2022.

Benjamin Grosof, Janine Bloomfield, Paul Fodor, Michael Kifer, Isaac Grosof, Miguel Calejo, and Theresa Swift. "Automated Decision Support for Financial Regulatory/Policy Compliance, using Textual Rulelog." *RuleML 2015.* Berlin, Germany. 2015.

PAPERS UNDER SUBMISSION

Heyuan Yao, **Willow Kowalik**, and Izzy Grosof. "Throughput-Optimal Multiresource-Job Scheduling with Continuous Requirement Distribution". Submitted, awaiting initial review.

Daniel Hurtado-Lange and Izzy Grosof. "Markov Modulated JSQ in Heavy Traffic Via the Poisson Equation".
Received major revision, revising.

Ziyuan Wang and Isaac Grosof. "Novel Lower Bounds on M/G/k Scheduling".
Received major revision, revising.

Yashaswini Murthy, Isaac Grosof, Siva Theja Maguluri, R. Srikant. "Performance of NPG in Countable State-Space Average-Cost RL".
Received major revision, resubmitted.

Runhan Xie, Ziv Scully, Isaac Grosof. "Optimal Multiserver Scheduling under General Service Constraints in Heavy Traffic".
Submitted, awaiting initial review.

TECHNICAL REPORTS & OTHER NONREFEREED PUBLICATIONS

Daniela Chang Kim, Izzy Grosof. "Queueing Sim." An open-source discrete-event simulator for single-server queueing policies. `https://crates.io/crates/queueing-sim`.

Isaac Grosof, Yige Hong, Mor Harchol-Balter. "Analysis of Markovian Arrivals and Service with Applications to Intermittent Overload." October 2024.

Isaac Grosof. "Optimal Scheduling in Modern Queueing Systems." *Thesis*. July 2023.

Isaac Grosof. "Open Problem - M/G/k/SRPT Under Medium Load." *Stochastic Systems*. Sep. 2019.

TEACHING

Winter 2026: Teaching IEMS 464, Advanced Queueing Theory. Second-year graduate-level course, advanced topics in queueing theory and scheduling theory. Quality of instruction rated as 6/6 (2026).

Winter 2025, '26: Teaching IEMS 460-1, Stochastic Processes I. First-year graduate-level course, proof-based look at Markov chains and Poisson processes. Quality of instruction rated as 5.75/6 (2025), 5.39/6 (2026). Student review: "This course was amazing! I learned a lot and the instructor clearly knew what they were doing and organized the class really well, which definitely added to my comprehension of the course."
2025 INFORMS Graduate Teaching Award from the IEMS Department at Northwestern.

Fall 2024, '25: Taught IEMS 315, Stochastic Models. Junior-senior level undergrad course, covering Markov chains and Poisson processes. Quality of instruction rated as 5.77/6 (2024), 5.40/6 (2025). Student review: "Izzy is one of the best instructors I have had at Northwestern. She is extremely passionate for teaching and cares that her students have a good grasp on the content."
Recipient of the 2025 Cole-Higgins Award for Excellence in Teaching from the McCormick School of Engineering at Northwestern.
Nominated for the Northwestern Canvas Hall of Fame, for Best Use of Lecture Videos/Recordings.

Fall 2020: TA for CMU 15-850, Advanced Algorithms (graduate level), assisting Prof. Anupam Gupta.

Fall 2019: TA for CMU 15-857, Analytical Performance Modeling & Design of Computer Systems (graduate level), assisting Prof. Mor Harchol-Balter.

Fall 2016 & Spring 2017: TA for MIT 6.1220 (MIT 6.046 at the time), Design and Analysis of Algorithms (undergraduate level), assisting Profs. Shafi Goldwasser, Nir Shavit, & Vinod Vaikunatanathan in fall 2016 and Profs. Debayan Gupta, Aleksander Madry & Bruce Tidor in spring 2017.

Fall 2015: Lab assistant for MIT 6.1010 (MIT 6.S04 at the time, 6.009 between), Fundamentals of Programming (undergraduate level), assisting Profs. Adam Chipala and Srini Devadas.

EXTERNAL SERVICE

Applied Probability Trust Editorial Board Member, 2026-present (Advances in Applied Probability & Journal of Applied Probability)
Co-organizer of Young Europeans in Queueing Theory 2025 Workshop.

Co-publications chair for ACM SIGMETRICS 2025.
TPC member for ACM SIGMETRICS/IFIP Performance 2024, '25, '26.
Reviewed papers for ACM SIGMETRICS, IFIP Performance, Management Science, and IEEE/ACM Transactions on Networking, ITCS, INFORMS Journal on Computing, ACM Performance Evaluation Review, Stochastic Systems, Operations Research, Queueing Systems.

DEPARTMENTAL SERVICE

2026: IEMS Seminars co-organizer.

Winter 2025: IEMS Graduate Committee member, proposing changes to graduate curiculum to increase student flexibility.

PRE-FACULTY LEADERSHIP & SERVICE

2019 - 2023: President of the CMU Humanist League, a philosophical discussion group devoted to the ideal of discourse over dogma.

2022 - 2023: Founding member of CMU CSD PhD Student Council, a student group which organizes social bonding events within the department.

2021 - 2023: Organizer of CSD weekly board games & socializing dinner event.

2021: Organizer of Computer Science Department (CSD) Introductory Course, which introduces new PhD students to the department.

2020 - 2021: CSD Admissions Committee member for the Theory group, reviewed ~ 200 applications.

Fall 2020: Organizer of CSD Theory Lunch Seminar series.

2020: Led events for CSD Open House.

2016: President of Random Hall, an MIT dorm of 93 residents.

OTHER EMPLOYMENT

Summer 2019: Facebook, Menlo Park, CA.
- Research Intern to develop a machine-learning-based SSD admission policy for Facebook's Tao caching architecture.
- Improved the tradeoff between hit ratio and SSD write rate.

Summer 2018: Microsoft Research, Seattle, WA.
- Research Intern to develop novel FPGA algorithms for linear algebra and solving linear programs.

Summer 2016: Jane Street Capital, LLC, New York City, NY.
- Software developer for non-obtrusive data collection about in-house trading.
- Software developer responsible for updating trading simulation package to accommodate new trade specification format.

Summer 2015: Coherent Knowledge, Seattle, WA.
- Knowledge Engineer to build demonstrations for the financial and natural language domains using the declarative logic programming language Ergo.

2013 - 2014: MIT Undergraduate Research Opportunities Program, Cambridge, MA.
- Researcher in Complexity Theory proving computational hardness of block puzzles and related agent motion problems.

Summer 2014: EMC Isilon, Seattle, WA.
- Software developer to replace the previous ad-hoc build platform with a modern Jenkins-based build platform.

TALKS GIVEN

Multiserver Stochastic Scheduling
- ECE Seminar, Northwestern, October 2024.
- Computer Science Seminar, Northwestern, December 2024.

Panel speaker on supporting trans researchers. October 2024.

New Lower Bounds on Optimal M/G/k Scheduling
- INFORMS Annual Meeting, October 2024.
- MAMA Workshop, ACM SIGMETRICS, June 2024.

Natural Policy Gradient for Queues
- Reinforcement Learning for Stochastic Networks, June 2024.

Analyzing Queues with Markovian Arrivals and Markovian Service
- **Tutorial** at ACM SIGMETRICS, June 2024.
- TU Eindhoven Seminar, June 2024.

The RESET and MARC Techniques for Multiserver-job Analysis
- IFIP Performance, November 2023.
- INFORMS Annual Meeting, October 2023.
- ACM SIGMETRICS Student Research Competition, June 2023.

Relative Arrivals: A New Drift Method for Time-Varying Arrivals
- Northwestern University, November 2023.
- SNAPP online seminar, October 2023.

Optimal Scheduling in Multiserver Queues (Thesis)
- Georgia Tech ISyE Student Seminar, September 2023.
- CMU: **Thesis Defense**, July 2023.

Multiserver Stochastic Scheduling: Analysis and Optimality (Job talk)
- Northwestern University, January 2023.

Optimal Scheduling in the Multiserver-job Model
- SIGMETRICS 2023.
- Harvard Theory Seminar, September 2022, hosted by Prof. Michael Mitzenmacher.
- MIT Algorithms & Complexity Seminar, September 2022, hosted by Prof. Ronitt Rubenfeld.

The RESET Technique for Multiserver-Job Analysis
- SIGMETRICS 2023, Student Research Competition.

New Stability Results for Multiserver-job Models via Product-form Saturated Systems
- MAMA workshop at SIGMETRICS 2023.

Stochastic Scheduling with Predictions
- STOC 2022, invited speaker in the Algorithms with Predictions workshop.

Work-Conserving Finite-Skip Models
- CMU Theory Lunch Seminar, January 2022.
- MIT LIDS Seminar, March 2022, hosted by Prof. Eytan Modiano.
- CORS/INFORMS International Meeting 2022

Multiserver-Job Systems
- INFORMS Annual Meeting 2021.

Nudge: Stochastically Improving upon FCFS
- SIGMETRICS 2021

Asymptotically Optimal Multiserver Scheduling
- UW Theory Seminar, March 2021, hosted by Prof. Anna Karlin.

- YEQT 2021, invited speaker.

Stability for Two-class Multiserver-job Systems
- CMU Theory Lunch Seminar, October 2020.

Load Balancing Guardrails: Keeping Your Heavy Traffic on the Road to Low Response Times
- SIGMETRICS 2019
- INFORMS Annual Meeting 2019
- Ohio State University Seminar, January 2020, hosted by Prof. Ness Shroff

M/G/k/SRPT under Medium Load
- INFORMS Annual Meeting 2018, Open Problem Session.

SRPT for Multiserver Systems
- MAMA workshop at SIGMETRICS 2018
- CMU Theory Lunch Seminar, September 2018.
- INFORMS Annual Meeting 2018
- Columbia Seminar, November 2018, hosted by Prof. Augustin Chaintreau
- Performance 2018

PROJECTS

2018 - present Programmatically-generated artwork:
`https://isaacg1.github.io/2018/12/06/programmatically-generated-artwork.html`

2014 - 2017: Author of new programming language: *Pyth*
Pyth is one of the best programming languages for code golfing, i.e. solving tasks with the shortest possible programs. Available at `https://github.com/isaacg1/pyth`